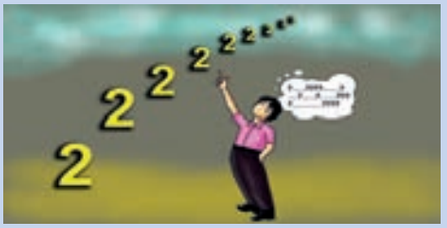
Think

We learnt that $2^2 = 2 \times 2$.

What is the value of 2^{2^2} ? Discuss.



3.3.4. Exponent Numbers with Different Base and Same Power

1. To understand the multiplication of exponent numbers with different base and same powers, let us consider the following example,

$$10^5 = 10 \times 10 \times 10 \times 10 \times 10$$
$$= (2 \times 5) \times (2 \times 5) \times (2 \times 5) \times (2 \times 5) \times (2 \times 5)$$
$$= (2 \times 2 \times 2 \times 2 \times 2) \times (5 \times 5 \times 5 \times 5 \times 5)$$
$$10^5 = 2^5 \times 5^5$$

But we know that, 10 = 2 × 5. Hence $10^5 = (2 \times 5)^5 = 2^5 \times 5^5$.

In general, for any non-zero integers 'a' and 'b' and for whole number 'm' $(m > 0)$,

$$a^m \times b^m = a \times a \times a \times \ldots \times a \ (m \text{ times}) \times b \times b \times b \times \ldots \times b \ (m \text{ times})$$
$$= (a \times b) \times (a \times b) \times (a \times b) \times \ldots \times (a \times b) \ (m \text{ times}) = (a \times b)^m$$

Therefore, $a^m \times b^m = (a \times b)^m$.

2. To understand the division of exponent numbers with different base and same powers, let us consider the follwing example,

$$10^5 = 10 \times 10 \times 10 \times 10 \times 10$$
$$= \left(\frac{20}{2}\right) \times \left(\frac{20}{2}\right) \times \left(\frac{20}{2}\right) \times \left(\frac{20}{2}\right) \times \left(\frac{20}{2}\right)$$
$$= \frac{20 \times 20 \times 20 \times 20 \times 20}{2 \times 2 \times 2 \times 2 \times 2}$$

Therefore, $10^5 = \frac{20^5}{2^5}$. But we know that, $10 = \left(\frac{20}{2}\right)$.

Hence, $10^5 = \left(\frac{20}{2}\right)^5 = \frac{20^5}{2^5}$.

Hence, for any two non-zero integers 'a' and 'b' and a whole number 'm' $(m > 0)$,

$$\left(\frac{a}{b}\right)^m = \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \times \ldots \times \left(\frac{a}{b}\right) \ (m \text{ times})$$
$$= \frac{a \times a \times a \times \ldots \times a\,(m \text{ times})}{b \times b \times b \times \ldots \times b\,(m \text{ times})} = \frac{a^m}{b^m}$$

Therefore, $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$.



Fig. 3.6

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